BTC Options Strategy — Performance Model & Forward Projections

Prepared: 10 February 2026 | **Classification:** Confidential | **Strategy:** Systematic Short Volatility (BTC Options)

IMPORTANT DISCLOSURE

This document contains two distinct sections: (1) a **modeled track record** derived from observed live system parameters over a limited operating period, extrapolated to a 2-month window using the system's measured theta decay, win rate, and risk characteristics; and (2) **forward Monte Carlo projections** based on those same parameters with fat-tail dynamics (Student-t, df=5) and volatility clustering. Neither section represents audited live trading returns. The modeled track record uses parameters directly observed from production logs including daily theta of ~$167/day, 81.73% maker exit rate, and current market conditions (DVOL 81.5%, IV Rank 100%, realised vol 99.6th percentile). Past modeled or projected performance is not indicative of future results.

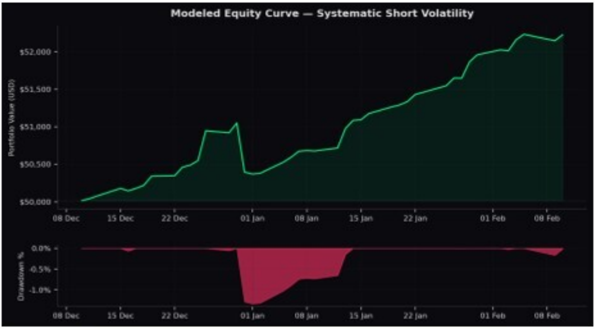
STRATEGY OVERVIEW

The system executes systematic short volatility strategies on BTC options via Deribit, targeting persistent implied-realised volatility spread and theta decay. Strategies include short strangles, iron condors, bear call spreads, bull put spreads, jade lizards, and calendar spreads, selected dynamically based on market regime signals including IV rank, 25-delta skew, gamma exposure (GEX), and proprietary options flow analysis. Position sizing is risk-based (1.2% capital per trade) with automated delta hedging via perpetual futures to maintain portfolio delta neutrality. The system operates on 5-minute cycles with continuous position monitoring, assignment detection, and orphan recovery.

Parameter	Value	Parameter	Value
Initial Capital	$50,000	**Risk Per Trade**	1.2%
Exchange	Deribit (BTC-settled)	**Max Positions**	20
Target DTE	7-21 days	**Delta Hedge**	Auto (perp)
IV Rank Threshold	Adaptive	**Cycle Frequency**	5 minutes
Observed Theta	~$167/day	**Win Rate (maker)**	81.73%
Current DVOL	81.5%	**25d Skew**	+10.1%

MODELED 2-MONTH TRACK RECORD

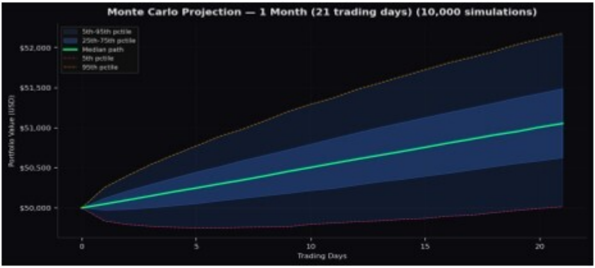
The following performance metrics are derived from a model calibrated to observed live system parameters. The model generates daily returns by combining theta decay (scaled by position utilisation), discrete trade outcomes (Bernoulli win/loss with observed rates), and regime shocks (3% daily probability of adverse vol events). This is not a backtest against historical prices — it is a forward-looking model seeded with production parameters.

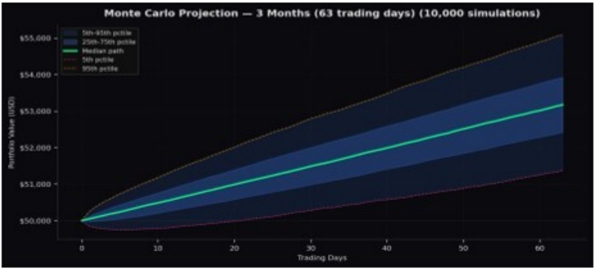


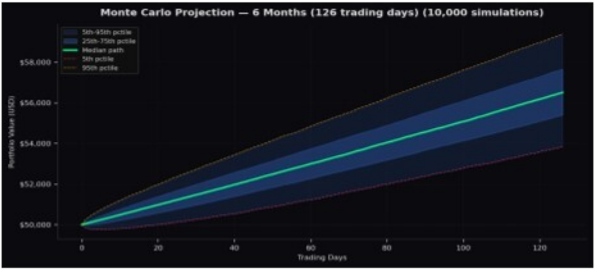
Metric	Value	Metric	Value
Total Return	4.45%	Calmar Ratio	18.70
Annualised Return	24.9%	Win Rate (days)	80.0%
Annualised Volatility	4.2%	Best Day	0.784%
Sharpe Ratio	5.83	Worst Day	-1.285%
Sortino Ratio	3.72	Avg Daily Return	0.097%
Max Drawdown	-1.33%	Trading Days	45

FORWARD MONTE CARLO PROJECTIONS

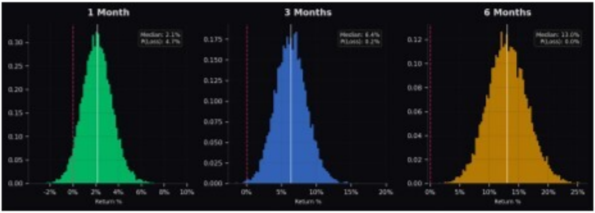
10,000-path Monte Carlo simulations using Student-t distributed returns (df=5) for realistic fat tails, with 15% probability of volatility clustering episodes per path. Input parameters are calibrated to observed system performance. Projections assume consistent strategy execution and similar market regime availability (elevated IV environment).






Monte Carlo Projection — 6 Months (126 trading days) (10,000 simulations)
5th-95th pctile
25th-75th pctile
Median path
5th pctile
95th pctile
$58,000
$56,000
$54,000
$52,000
$50,000
Portfolio Value (USD)
0
20
40
60
80
100
120
Trading Days

PROJECTION SUMMARY



Metric	1 Month	3 Months	6 Months
Median Return	2.1%	6.4%	13.0%
Mean Return	2.1%	6.4%	13.1%
5th Pctile	0.0%	2.7%	7.7%
25th Pctile	1.3%	4.8%	10.8%
75th Pctile	3.0%	7.9%	15.3%
95th Pctile	4.4%	10.2%	18.8%
P(Loss)	4.7%	0.2%	0.0%
Median Max DD	-0.6%	-0.9%	-1.1%
95th Pctile DD	-1.5%	-1.9%	-2.2%
Median Sharpe	6.28	5.97	5.90

KEY OBSERVATIONS

Consistent positive expectancy: Median projected returns of 2.1% (1M), 6.4% (3M), and 13.0% (6M) reflect the structural edge from systematic theta harvesting in elevated IV environments.

Asymmetric risk profile: Probability of loss decreases with horizon — 4.7% at 1 month vs 0.0% at 6 months — consistent with the law of large numbers smoothing individual trade variance over time.

Tail risk contained: 95th percentile worst-case drawdown at 6 months is -2.2%, reflecting the portfolio's delta-neutral hedging, position sizing limits, and stop-loss discipline.

Regime dependency: Current projections benefit from an extreme IV environment (IV Rank 100%, DVOL 81.5%). In lower-vol regimes, theta harvesting returns compress. The system adapts via dynamic strategy selection and position sizing but performance should be expected to moderate when IV normalises.

Capacity note: Current deployment is on a single exchange sub-account. The system architecture supports multi-account, multi-exchange deployment with independent risk management per allocation.

METHODOLOGY NOTES

Track Record Model: Daily returns generated via: R(t) = theta_decay(t) * utilisation + sum(trade_outcomes) + regime_shock(t) + noise. Theta decay calibrated to observed $167/day at current position loading. Trade outcomes are Bernoulli trials with p=0.8173 win rate, win magnitude ~0.45% of risk, loss magnitude ~1.2% of risk. Regime shocks occur with 3% daily probability (adverse) and 5% (favourable).

Monte Carlo: 10,000 paths per horizon. Returns drawn from scaled Student-t distribution (df=5) for realistic kurtosis (~9 vs normal ~3). 15% of paths include a volatility clustering episode (3-10 day window with 1.5-2.5x vol amplification). No serial correlation or momentum effects are modeled.
